



Finding the original amount after a decrease

Task A: Finding the original value

- 1) Using your preferred method, answer the following questions. Give your answers to an appropriate degree of accuracy.

- a) A t-shirt costs £21 in a sale offering a 25% discount.

What is the cost of the t-shirt before the discount?



$$\times \frac{100}{75}$$

%	price (£)
75	21
100	28

$$\times \frac{100}{75}$$

The price of the t-shirt before the discount is £28

- b) A car's value depreciates by 22% in the first year. It is now worth £11 544. What was the cost of the car originally?



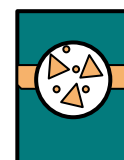
$$\times \frac{100}{78}$$

%	cost (£)
78	11 544
100	14 800

$$\times \frac{100}{78}$$

The cost of the car was originally £14 800

- c) A packet of crisps contains 33% less fat. This low fat packet contains 5.3 g of fat. How much fat is in a regular packet?



$$\times \frac{100}{67}$$

%	grams
67	5.3
100	7.9

$$\times \frac{100}{67}$$

The regular packet contains 7.9 g of fat.



Task B: Using a multiplier

- 1) Given the new amount and percentage decrease find the original amount. All of the answers are in the grid except for one.

Write a percentage question that gives this answer.

#	Percentage Decrease	New Amount
a	10%	315
b	15%	289
c	45%	176
d	12%	306.24
e	36%	245.76
f	80%	73.6
g	12.5%	288.75
h	3.6%	303.66
i	<i>E.g. 35%</i>	<i>211.25</i>

c 320	e 384	b 340
h 315	325	a 350
f 368	g 330	d 348

Task C: Mixed reverse percentage problems

- 1) Decide whether each of the following questions indicate a decrease or increase question.

a) A watch is sold for £46.80, this was a **loss** of 35%.

decrease

b) A bag of apples contains an **extra** 25% free. The bag contains 20 apples.

increase

c) The value of a motorbike **depreciates** by 20%. It is now worth £6820.

decrease

d) A savings account earns 2.2% **interest**. The account now contains £613.20

increase

e) You buy a laptop for £244.40, this price has been **discounted** by 6%.

decrease



2) Using your preferred method, find the original amounts in the following questions.

a) A watch is sold for £46.80. This was a loss of 35%.

$$46.80 \div 0.65 = £72$$

b) A bag of apples contains an extra 25% free. The bag contains 20 apples.

$$20 \div 1.25 = 16 \text{ apples}$$

c) The value of a motorbike depreciates by 20%. It is now worth £6820

$$6820 \div 0.8 = £8525$$

d) A savings account earns 2.2% interest. The account now contains £613.20

$$613.20 \div 1.022 = £600$$

e) You buy a laptop for £244.40. This price has been discounted by 6%.

$$244.40 \div 0.94 = £260$$

